

In mathematics, statements are typically of the form "If A, then B."

We must verify the validity of math statements through proof.

To prove math statements, we use axioms, prior theorems, and established rules.

Example of an axiom: In Euclidean geometry, there exists parallel lines.

Here's what a proof might look like:

Statement/claim: If A, then B.

Proof. Assume A.

Axioms, Thms, rule. {
Then (consequence of A)
Thus (another consequence)
:
Then B.

Another useful way to prove "If A, then B" is proof by contradiction.

Claim: If A, then B.

Proof. Assume A but not B (i.e. A is true & B is false).

Then..... (consequence of A not B)

Then
⋮

Ridiculousness, e.g. $1 = 2$.

This means our initial assumption that A true, but B is not, is false! Thus, original claim "If A, then B" must be true.

Another type of logical statement:

A exists.

Proof: (list of things we already know).
magic..... out pops A ☺

Another way to prove :

If A , then B .

Proof. (Contrapositive) Assume not B .
(B not true).

Then ----

Then ----

⋮

Then, A is not true.

1.2. Basic Set Theory Definitions

Def. A set is an unordered collection of distinct objects, which are called elements or members of the set.

If x is an element of a set S , we write $x \in S$.


"x is in S"
"belongs to"

Ex. Take the set $\mathbb{N}$ of natural numbers
 $1, 2, 3, 4, 5, \dots$ Then $53 \in \mathbb{N}$.

But $-2 \notin \mathbb{N}$.


"does not belong to"

Set-builder notation:

$$S = \{ \text{elements} : \text{conditions used to generate elements} \}$$

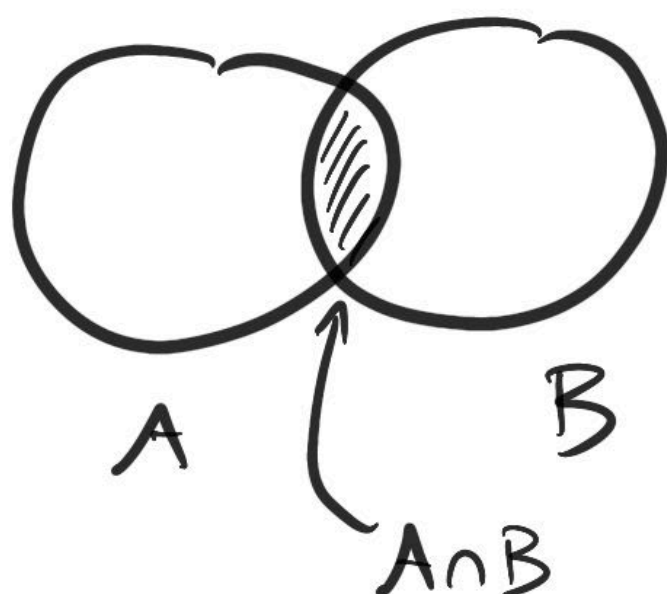
Ex. The rational numbers:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

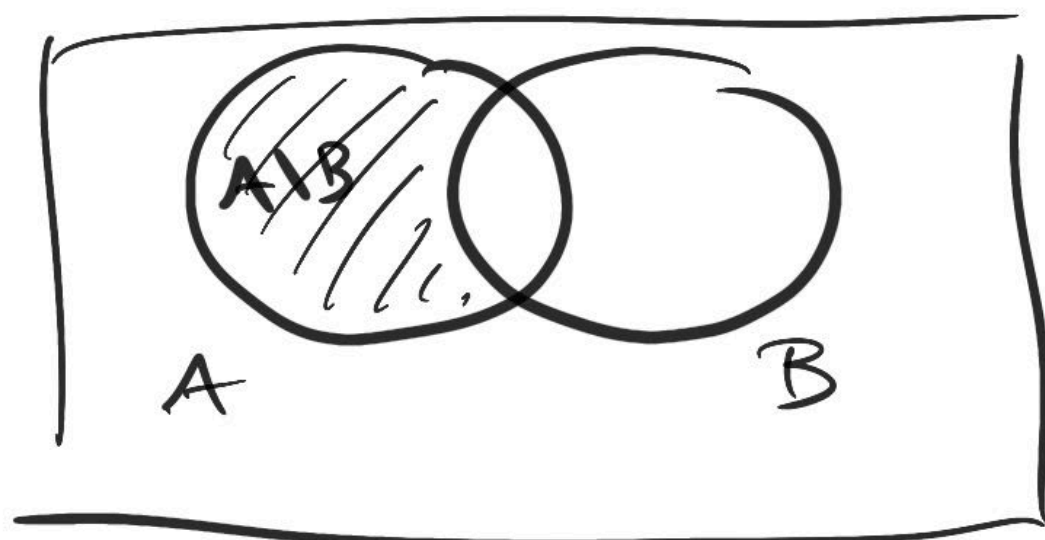

integers.

The intersection of A and B is the set

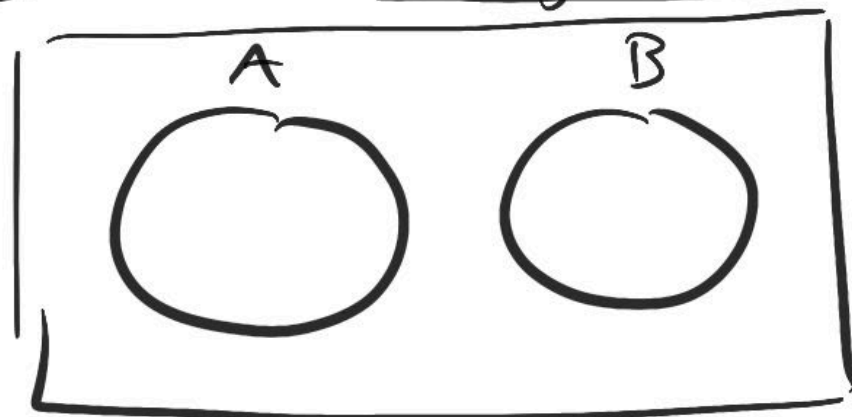
$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$



Set difference : $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$



A and B are disjoint if $A \cap B = \emptyset$.



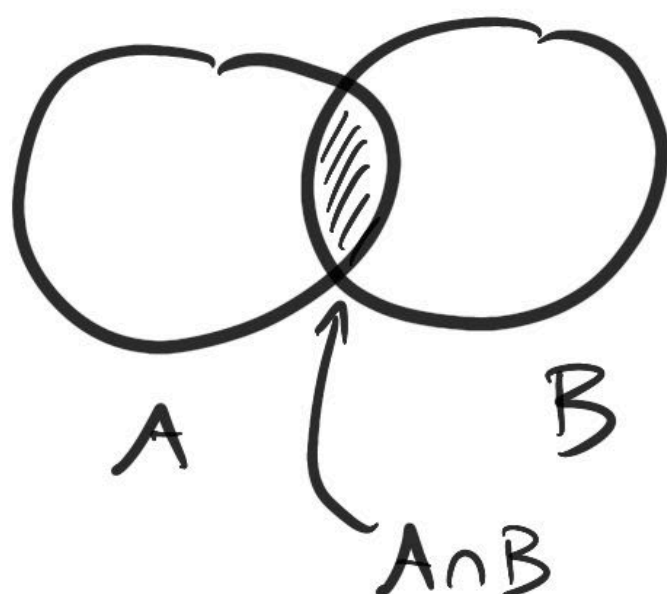
If $A \subseteq U$ for a universal set U (typically $U = \mathbb{R}$), the complement of A in U is

$$A^c = U \setminus A$$

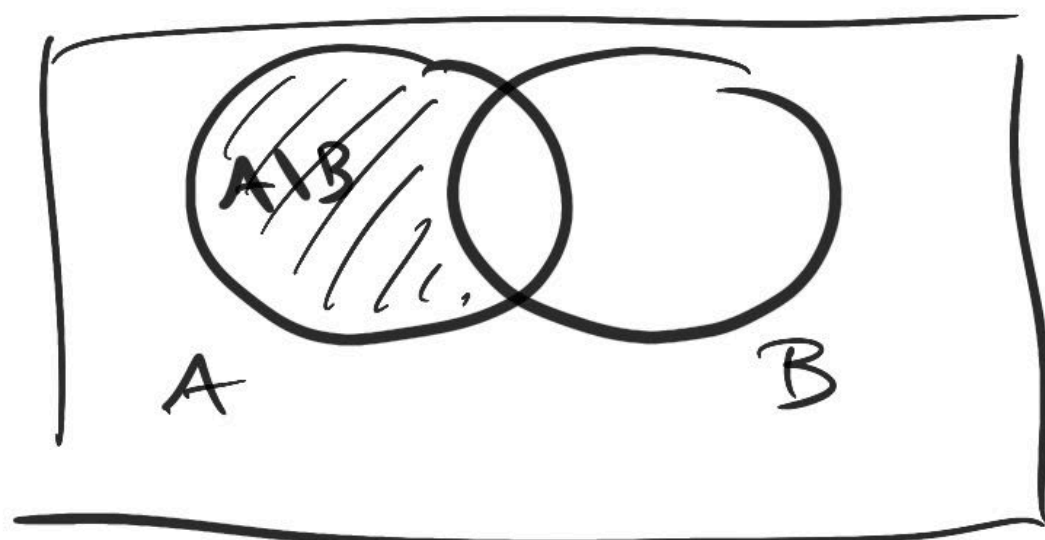


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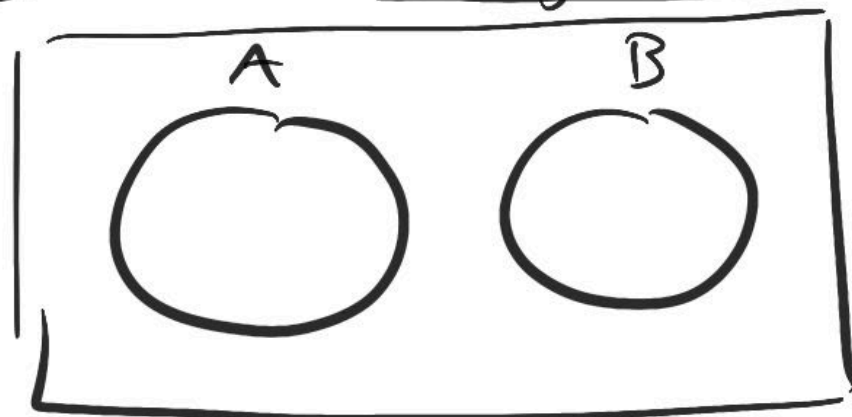
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If $A \subseteq U$ for a universal set U (typically $U = \mathbb{R}$), the complement of A in U is

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Def. The Cartesian product of A, B is

$$A \times B = \{ (a, b) : a \in A, b \in B \}.$$

e.g. $\mathbb{R} \times \mathbb{R} = \{ (x, y) : x, y \in \mathbb{R} \} = \mathbb{R}^2$

If $A_1, A_2, A_3, \dots, A_n$ are sets, the Cartesian product of them is

$$A_1 \times A_2 \times A_3 \times \dots \times A_n$$

$$= \prod_{i=1}^n A_i = \{ (a_1, a_2, a_3, \dots, a_n) : a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n \}.$$

Also, the union of $A_1, \dots, A_n$ is

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n.$$

Likewise,

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n.$$

If $A_1, A_2, A_3, \dots$ an infinite collection of sets, then:

$$\bullet \quad \prod_{i=1}^{\infty} A_i = A_1 \times A_2 \times A_3 \times \dots$$

$$\bullet \quad \bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots$$

$$\bullet \quad \bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap A_3 \cap \dots$$

Def. The cardinality of a set A is the "number" of elements in A , denoted $|A|$, or $\#A$.

Ex. $A = \{1, \odot, \square\}$

$$|A| = 3$$

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$$|\mathbb{N}| = ? \quad |\mathbb{R}| = ? \quad |\mathbb{Q}| = ?$$

Def. For a set A , the power set $\mathcal{P}(A)$ is the set of all subsets of A :

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Ex. $A = \{1, \odot, \Delta\}$

$$\mathcal{P}(A) = \left\{ \{1\}, \{\odot\}, \{\Delta\}, \{1, \odot\}, \{1, \Delta\}, \{\odot, \Delta\}, \{1, \odot, \Delta\}, \emptyset \right\}$$

$$|\mathcal{P}(A)| = 8 \quad (|A| = 3).$$

Exercise: Given a finite set A of n elements (i.e. $|A| = n$), find $|\mathcal{P}(A)|$.

Def. Given two sets A and B , suppose that each element $x \in A$ is associated, in some way, to exactly one element of B , call this element $f(x) \in B$. Then f is called a function from A to B , written $f: A \rightarrow B$.

e.g. $f: \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = x^2 \in \mathbb{N}$.

Here, A is the domain, B the codomain.

(in our example, $3 \in \mathbb{N}$ is in the codomain, but there's no $x \in \mathbb{N}$ with $x^2 = 3$, $f(x) \neq 3$).

The set $\{f(x) : x \in A\}$ is the range of f .

$$f(A) = \{f(x) : x \in A\} \subseteq B.$$

$\text{ran}(A)$

$\text{range}(A)$

Def. A function $f: A \rightarrow B$ is injective or one-to-one if $f(a) = f(b)$ implies $a = b$.

same statement! $\left\{ \begin{array}{l} \cdot \text{ If } f(a) = f(b), \text{ then } a = b. \checkmark \\ \cdot \text{ If } a \neq b, \text{ then } f(a) \neq f(b). \end{array} \right.$

Proposition. $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$ is injective.

Proof. Assume $f(x) = f(y)$. Then $x^3 = y^3$. Cube root both sides:

$$\sqrt[3]{x^3} = \sqrt[3]{y^3} \Rightarrow x = y. \text{ Thus,}$$

$f(x) = f(y) \Rightarrow x = y$, whence f is injective. $\square$

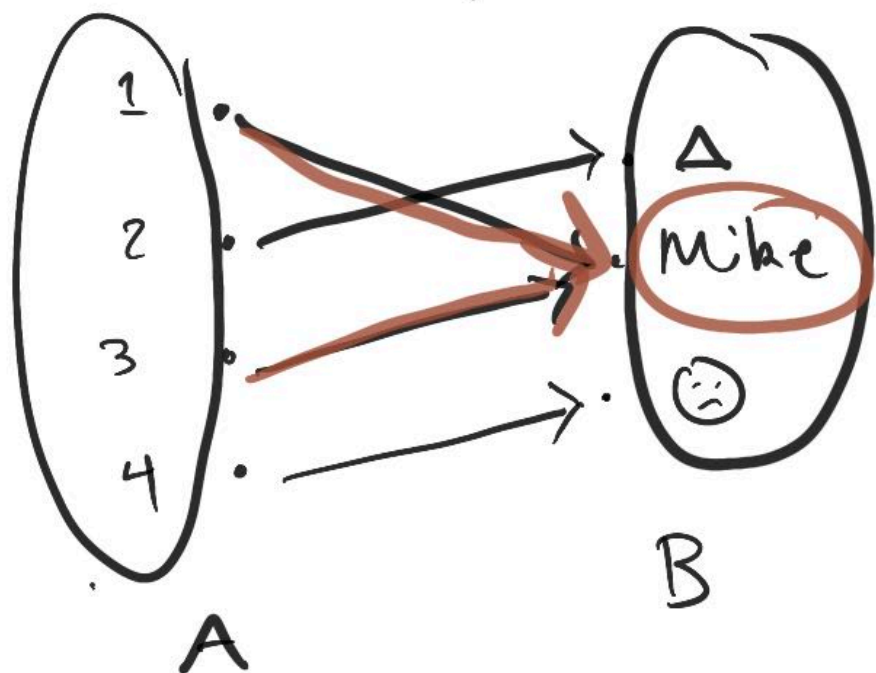
Def. $f: A \rightarrow B$ is surjective or onto if for every $b \in B$, there exists some $a \in A$ for which $f(a) = b$.

i.e. $f(A) = B$.

Def. $f: A \rightarrow B$ is bijjective if it is both injective and surjective.

Ex. $A = \{1, 2, 3, 4\}$, $B = \{\Delta, \text{Mike}, \text{☹}\}$

f



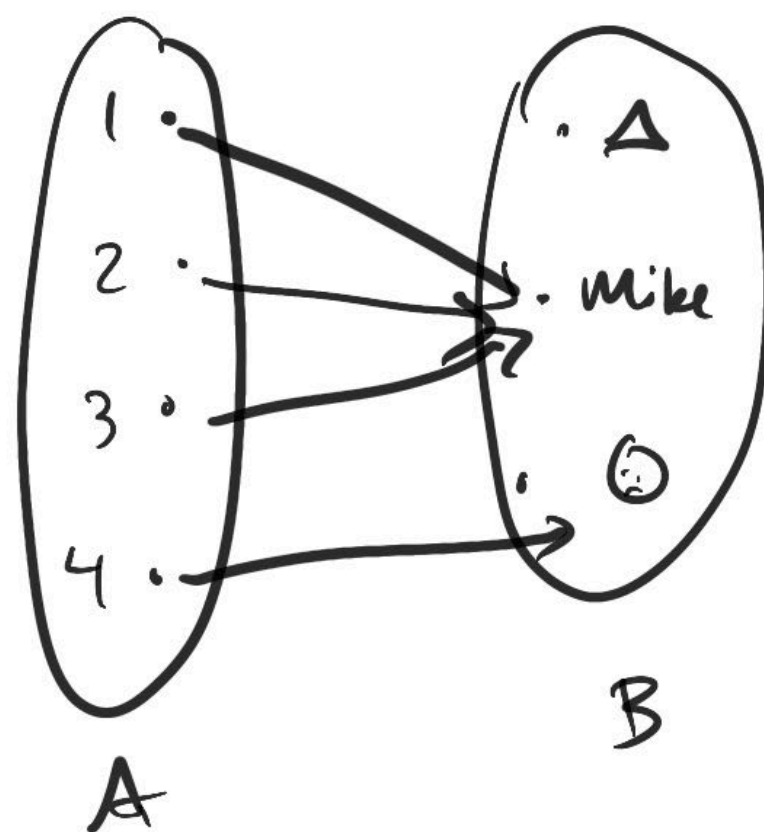
$f(1) = f(3)$,
but $1 \neq 3$

f is surjective!!

(arrow pointing to everything in B).

f is not injective!!

g



g Not surjective.

Δ is not a target,

$\Delta \neq f(x), x \in A$.

Not injective.

Number Systems.

- Natural numbers $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- Integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$
- Rationals $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$.

To solve $2x = 3$, we need to be in $\mathbb{Q}$
solution: $x = 3/2$.

$\nabla!$ $1 \in \mathbb{Q}$, $0 \in \mathbb{Q}$, $-1000000 \in \mathbb{Q}$
 $\uparrow$ $1 = \frac{1}{1} = \frac{3}{3}$ $\frac{0}{3}$

So far, we have nested sets:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

To solve $x^2 = 2$, $\mathbb{Q}$ does n't suffice!!

We know $x = \pm \sqrt{2} \notin \mathbb{Q}$

Solving equations like this gives us the set $\mathbb{R}$ of real numbers.

$$\mathbb{R} = \left\{ \lim_{n \rightarrow \infty} q_n : (q_n) \text{ sequence in } \mathbb{Q} \right\}$$

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \boxed{\mathbb{R}} \subseteq \mathbb{C}$$

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}, \quad i = \sqrt{-1}.$$

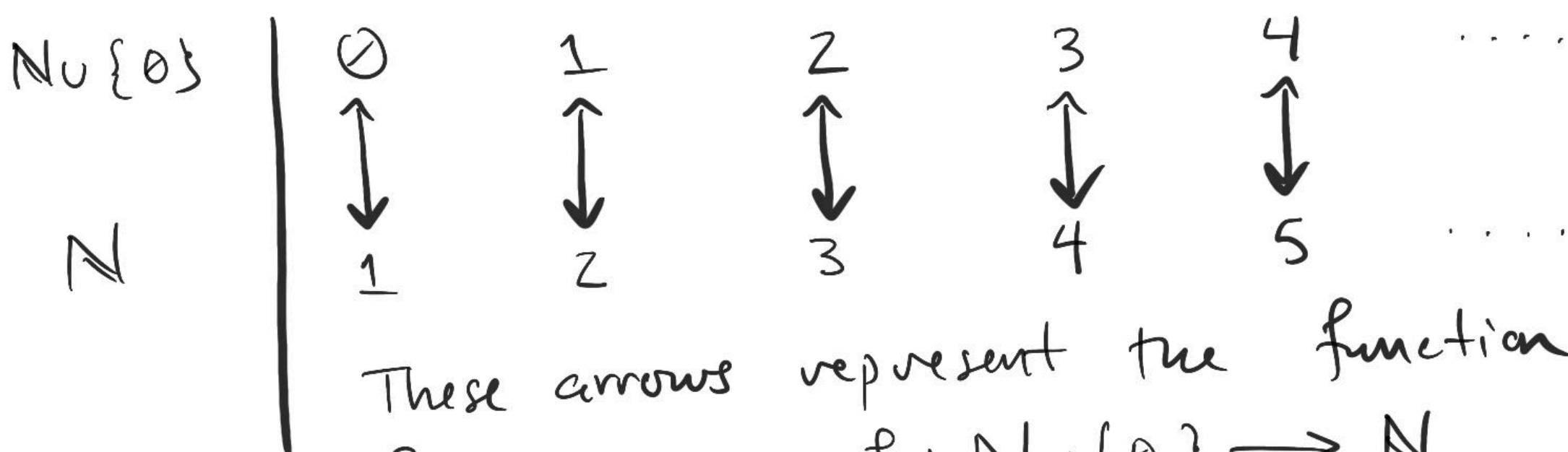
(Back to size or cardinality:)

$$A = \{1, 2, 3, 4, 5\}, \quad |A| = 5$$

Notation: $|\mathbb{N}| = \aleph_0$ "aleph null"
means "smallest infinity".

($\aleph_0$ is an ex. of a cardinal number).

$$\mathbb{N} \cup \{0\} = \{0, 1, 2, 3, \dots\}$$



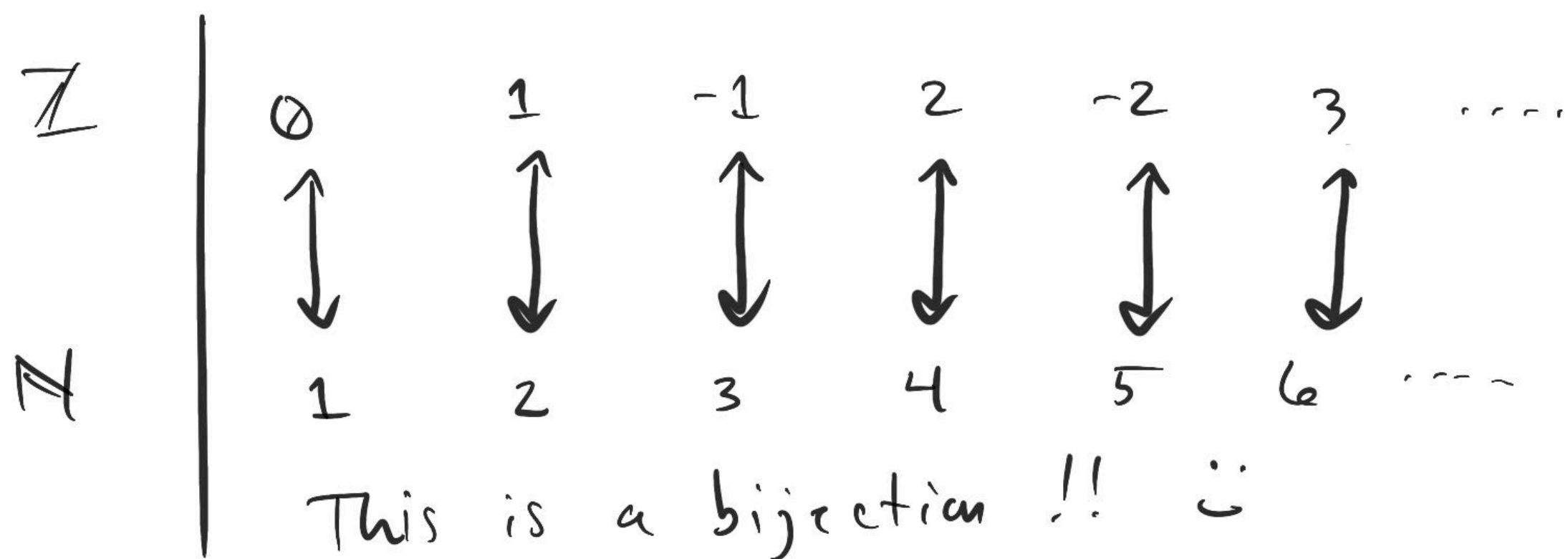
These arrows represent the function
 $f(x) = x + 1, \quad f: \mathbb{N} \cup \{0\} \rightarrow \mathbb{N}$

it's a bijection !!

$$\text{Thus, } |\mathbb{N} \cup \{0\}| = |\mathbb{N}|$$

Adding any number of elements to $\mathbb{N}$
does not effect cardinality (size) !!

Compare $|\mathbb{Z}|$ to $|\mathbb{N}| = \aleph_0$.



$$\text{So } |\mathbb{Z}| = |\mathbb{N}| = \aleph_0$$

Is $|\mathbb{Q}|$ bigger than $|\mathbb{N}|$?

Claim: $|\mathbb{Q}| = |\mathbb{N}|$.



Prove this!! Find a way to
count $\mathbb{Q}$ with a starting
point. Try to arrange $\mathbb{Q}$
in a way that allows to count
one by one !! Can also $|\mathbb{Q}| = |\mathbb{Z}|$.

It turns out that $|\mathbb{R}| > |\mathbb{N}| = \aleph_0$.

$\mathbb{R}$ is an uncountable set.

This is proven using Cantor's diagonal argument.

There are infinitely many types of infinity, $\aleph_0$ being the smallest.

Rationals

$\mathbb{Q}$ is nested: between any two rationals is another one. (in fact, inf. many).

$\mathbb{Q}$ is closed under addition, subtraction, multiplication, and division:

- If $p, q \in \mathbb{Q}$, then $p + q \in \mathbb{Q}$.
- If $p, q \in \mathbb{Q}$, then $p - q \in \mathbb{Q}$.
- If $p, q \in \mathbb{Q}$, then $p \cdot q \in \mathbb{Q}$.
- If $p, q \in \mathbb{Q}$, then $\frac{p}{q} \in \mathbb{Q}$ provided $q \neq 0$.

A set of numbers satisfying these properties is a field.

But $\mathbb{Q}$ has some problems....

- It's possible that $p \in \mathbb{Q}$ but yet $\sqrt{p} \notin \mathbb{Q}$.
- $\mathbb{Q}$ has "holes": It's possible that $p_1, p_2, p_3, \dots \in \mathbb{Q}$ (inf. many) but yet the sequence $p_1, p_2, p_3, \dots$ "converges" to some $p \notin \mathbb{Q}$.
- $\mathbb{Q}$ is not "algebraically closed": there are polynomials whose roots are not rational, e.g.

$$x^2 - 5.$$

- If we "fill in" these holes, we'll get a number system on which we can do calculus!!